A (mini) introduction to the p-adic integers

Nicolas Kress

Feb. 2026

Let p be a fixed prime number.

1 The p-adic absolute value

Given an integer $n \in \mathbb{Z}$, we define the **p-adic valuation of** n as:

$$\nu_p(n) := \sup\{k \geq 0 : p^k \text{ divides } n\} \in \bar{\mathbb{N}}$$

By convention, $\nu_p(0) := \infty$. This definition is extended to any rational $q = \frac{a}{b} \in \mathbb{Q}$ by defining:

$$\nu_p\left(\frac{a}{b}\right) := \nu_p(a) - \nu_p(b)$$

We can easily verify that the previous expression is defined independently of the representative $\frac{a}{b}$, as ν_p is trivially additive over $\mathbb{Z}$.

For any rational $x \in \mathbb{Q}$, we define the **p-adic absolute value of** x by:

$$|x|_p := p^{-\nu_p(x)}$$

with, by convention, $|0|_p := 0$.

Let's show that $|\cdot|_p : \mathbb{Q} \to \mathbb{Q}_+$ is an absolute value over $\mathbb{Q}$:

1. *(definite positivity)* For any $x \in \mathbb{Q}$ such that $|x|_p = 0$, we have $\nu_p(x) = \infty$, meaning that necessarily $x = 0$.

2. *(multiplicativity)* For any $x, y \in \mathbb{Q}$, since $\nu_p(xy) = \nu_p(x) + \nu_p(y)$, we have $|xy|_p = |x|_p |y|_p$.

3. *(triangle inequality)* To show the triangle inequality, we are going to exploit the **non-Archimedean** triangle inequality:

   $$|x + y|_p \leq \max\left(|x|_p, |y|_p\right)$$

   Equivalently, we are going to show:

   $$\nu_p(x + y) \geq \min(\nu_p(x), \nu_p(y))$$

   Let $x = \frac{a}{b},\ y = \frac{c}{d}$ be two rational numbers. Then:

   $$\nu_p\left(\frac{a}{b} + \frac{c}{d}\right) = \nu_p(ad + bc) - \nu_p(bd)$$

   But ad and bc are integers. Clearly, denoting $r := \min(\nu_p(ad), \nu_p(bc))$, p^r divides both ad and bc, meaning that it divides $ad + bc$. It follows that $r \leq \nu_p(ad + bc)$. So:

   $$\begin{aligned}\nu_p(x + y) &\geq \min\left(\nu_p(ad) - \nu_p(bd),\ \nu_p(bc) - \nu_p(bd)\right) \\ &= \min\left(\nu_p\left(\frac{a}{b}\right),\ \nu_p\left(\frac{c}{d}\right)\right) \\ &= \min(\nu_p(x), \nu_p(y))\end{aligned}$$

We have thus shown that $|\cdot|_p$ is a (non-Archimedean, or **ultrametric**) absolute value over $\mathbb{Q}$. Note that it is not complete. Consider for example the sequence:

$$\forall n \geq 0, \quad x_n := \sum_{k=0}^{n} p^{k^2}$$

It is easy to convince oneself that $(x_n)_{n\geq 0}$ is a Cauchy sequence, and yet, $(x_n)_{n\geq 0}$ does not converge in $\mathbb{Q}$. The study of the completion of $\mathbb{Q}$ with respect to $|\cdot|_p$ is thus justified.

2 The p-adic number field $\mathbb{Q}_p$

The completion of $\mathbb{Q}$ with respect to the p-adic absolute value $|\cdot|_p$ is denoted $\mathbb{Q}_p$, and is called the **p-adic number field**. Formally, $\mathbb{Q}_p$ is the quotient of the set $\mathcal{C}$ of Cauchy sequences over $\mathbb{Q}$ (w.r.t. $|\cdot|_p$) by the equivalence relation defined by:

$$(x_n) \sim (y_n) \iff \lim_{n\to\infty} (x_n - y_n) = 0$$

Let $x, y \in \mathbb{Q}_p$ and $(x_n)_{n\geq 0} \in x$, $(y_n)_{n\geq 0} \in y$. We define the sum of x and y (as elements of $\mathbb{Q}_p$) by:

$$x + y := \left[(x_n + y_n)_{n\geq 0}\right]$$

where $[(z_n)]$ denotes the class of (z_n) in $\mathcal{C}/_\sim =: \mathbb{Q}_p$. We similarly define the product. Since $\mathcal{C}$ is a $\mathbb{Q}$-algebra, all we have done is show that the algebraic operations descend onto the quotient set.

Let's now show that any non-zero $x = [(x_n)] \in \mathbb{Q}_p$ has an inverse. Since $x \neq 0$, $\lim x_n \neq 0$. There therefore exists $\varepsilon > 0$ and $n_0 \geq 0$ such that

$$\forall n \geq n_0, \quad |x_n|_p > \varepsilon$$

Notably, $x_n \neq 0$ for all $n \geq n_0$. Then define $y_n := x_n^{-1}$ for all $n \geq n_0$, and give y_n some arbitrary value for any $n < n_0$. We can show that $(y_n)_{n\geq 0}$ is a Cauchy sequence, and notably, we can denote $y := [(y_n)] \in \mathbb{Q}_p$. Considering the product xy, we have:

$$xy = \lim_{n\to\infty} x_n y_n = \lim_{n\to\infty} 1 = 1$$

So $\mathbb{Q}_p$ is a field. Moreover, we can extend the definition of the p-adic absolute value $|\cdot|_p$ to $\mathbb{Q}_p$, by:

$$|x|_p := \lim_{n\to\infty} |x_n|_p$$

$|\cdot|_p$ is still a non-Archimedean absolute value over $\mathbb{Q}_p$.

3 The p-adic integer ring $\mathbb{Z}_p$

The **p-adic integer set** $\mathbb{Z}_p$ is defined as the closed unit ball of $\mathbb{Q}_p$ with respect to $|\cdot|_p$. Since this absolute value is ultrametric, this unit ball is a sub-ring of $\mathbb{Q}_p$.

We can show that $p\mathbb{Z}_p$ is a maximal ideal of $\mathbb{Z}_p$, and the quotient $\mathbb{Z}_p/p\mathbb{Z}_p$ is isomorphic to the finite field $\mathbb{Z}/p\mathbb{Z}$. Before we reach the series representation theorem, we are going to exploit the following lemma:

Lemma 1. *For any $x \in \mathbb{Z}_p$ such that $|x|_p = 1$, there exists a unique $\alpha \in \{0, \ldots, p-1\}$ such that:*

$$|x - \alpha|_p < 1$$

Proof. Denote $\phi : \mathbb{Z}_p/p\mathbb{Z}_p \to \mathbb{Z}/p\mathbb{Z}$ the isomorphism previously discussed. Also denote $X \in \mathbb{Z}_p/p\mathbb{Z}_p$ the class of x. We can then define $\alpha := \phi(X) \in \{0, \ldots, p-1\}$. By identifying α to a p-adic number (as a rational number), we have $\alpha \in \mathbb{Z}_p$, and notably:

$$\phi([x - \alpha]) = \phi(X - [\alpha]) = \phi(X) - \phi([\alpha]) = 0$$

where $[x-\alpha]$ denotes the class of $x-\alpha$ in $\mathbb{Z}_p/p\mathbb{Z}_p$. Because ϕ is an isomorphism, we have $[x-\alpha]=0$, and so necessarily $x-\alpha \in p\mathbb{Z}_p$. It follows finally that $|x-\alpha|_p < 1$.

Suppose there exists $\beta \in \{0,\dots,p-1\}$ such that $|x-\beta|_p < 1$. Then $x-\beta \in p\mathbb{Z}_p$, meaning $[x-\beta]=0$ and subsequently $\alpha = \phi(X) = \phi([\beta]) = \beta$, hence uniqueness. □

Thanks to the previous result, we can finally show the **series representation** theorem, which fundamentally simplifies the way we think about p-adic integers.

Theorem 1 (Series representation)**.** *Let x be a p-adic integer. There exists a unique sequence $(a_k)_{k\geq 0}$ with values in $\{0,\dots,p-1\}$ such that:*

$$x = \sum_{k=0}^{\infty} a_k p^k$$

In other words,

$$\mathbb{Z}_p = \left\{ \sum_{k=0}^{\infty} a_k p^k \ : \ \forall k \geq 0,\ a_k \in \{0,\dots,p-1\} \right\}$$

Proof. Obtained by iterating the previous lemma. Only a sketch is given here. For some $x \in \mathbb{Z}_p$, the p-adic number $\frac{x}{|x|_p}$ has p-adic absolute value 1. By the previous lemma, we can thus find a_0 such that:

$$\Big| x - a_0 \, |x|_p \Big|_p < |x|_p$$

By iterating, we can construct a sequence of p-adic numbers that converge to x but that are all of the form $\sum_{k=0}^{N} a_k \, |x_k|_p$, which grants us the result. □